

Eigen Values and Vectors

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Question : What can we conclude if a matrix $\mathbf{A}$ is symmetric?

Proof :

Case 1 : $\mathbf{A}$ is real and symmetric

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a real symmetric matrix, i.e.,

$$\mathbf{A} = \mathbf{A}^\top \quad (1)$$

Suppose λ is an eigenvalue of $\mathbf{A}$ with eigenvector $\mathbf{v} \neq \mathbf{0}$:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \quad (2)$$

Taking the conjugate transpose of $\mathbf{v}$ and multiplying on the left:

$$\mathbf{v}^* \mathbf{A} \mathbf{v} = \lambda \mathbf{v}^* \mathbf{v} \quad (3)$$

Since $\mathbf{A}$ is real symmetric, $\mathbf{A} = \mathbf{A}^* = \mathbf{A}^\top$, therefore:

$$\mathbf{v}^* \mathbf{A} \mathbf{v} = (\mathbf{A}\mathbf{v})^* \mathbf{v} = (\lambda\mathbf{v})^* \mathbf{v} = \bar{\lambda} \mathbf{v}^* \mathbf{v} \quad (4)$$

Equating both expressions for $\mathbf{v}^* \mathbf{A} \mathbf{v}$:

$$\lambda \mathbf{v}^* \mathbf{v} = \bar{\lambda} \mathbf{v}^* \mathbf{v} \quad (5)$$

$$(\lambda - \bar{\lambda}) \mathbf{v}^* \mathbf{v} = 0 \quad (6)$$

Since $\mathbf{v} \neq \mathbf{0}$, $\mathbf{v}^* \mathbf{v} > 0$. Hence:

$$\lambda = \bar{\lambda} \quad (7)$$

Therefore, every eigenvalue λ of $\mathbf{A}$ is real.

Eigenvectors corresponding to distinct eigenvalues are orthogonal

Let

$$\mathbf{A}\mathbf{u} = \alpha\mathbf{u} \quad (8)$$

$$\mathbf{A}\mathbf{w} = \beta\mathbf{w} \quad (9)$$

Where $\alpha \neq \beta$ are distinct eigenvalues. Multiplying the first equation on the left by $\mathbf{w}^\top$:

$$\mathbf{w}^\top \mathbf{A} \mathbf{u} = \alpha \mathbf{w}^\top \mathbf{u} \quad (10)$$

Since $\mathbf{A}$ is symmetric, $\mathbf{A} = \mathbf{A}^\top$. Hence:

$$\mathbf{w}^\top \mathbf{A} \mathbf{u} = (\mathbf{A}\mathbf{w})^\top \mathbf{u} = (\beta\mathbf{w})^\top \mathbf{u} = \beta \mathbf{w}^\top \mathbf{u} \quad (11)$$

Equating both forms:

$$\alpha \mathbf{w}^\top \mathbf{u} = \beta \mathbf{w}^\top \mathbf{u} \quad (12)$$

Simplifying:

$$(\alpha - \beta) \mathbf{w}^\top \mathbf{u} = 0 \quad (13)$$

Since $\alpha \neq \beta$, it follows that:

$$\mathbf{w}^\top \mathbf{u} = 0 \quad (14)$$

Hence, eigenvectors corresponding to distinct eigenvalues are orthogonal.

Proving that Eigen Vector is real

Suppose $\lambda \in \mathbb{R}$ is an eigenvalue of $\mathbf{A}$ with a (possibly complex) eigenvector $\mathbf{v} \in \mathbb{C}^n$, satisfying

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \quad (15)$$

Considering the conjugate eigenvector

Since $\mathbf{A}$ is real, taking the complex conjugate of both sides gives

$$\mathbf{A}\bar{\mathbf{v}} = \bar{\lambda}\bar{\mathbf{v}} \quad (16)$$

But for a real symmetric matrix, $\lambda = \bar{\lambda}$ (as proven earlier). Hence,

$$\mathbf{A}\bar{\mathbf{v}} = \lambda\bar{\mathbf{v}} \quad (17)$$

Thus, $\bar{\mathbf{v}}$ is also an eigenvector corresponding to the same eigenvalue λ .

Constructing real vectors from $\mathbf{v}$ and $\bar{\mathbf{v}}$

Define

$$\mathbf{u} = \mathbf{v} + \bar{\mathbf{v}} \quad (18)$$

$$\mathbf{w} = i(\mathbf{v} - \bar{\mathbf{v}}) \quad (19)$$

Both $\mathbf{u}$ and $\mathbf{w}$ are **real vectors**, because $\mathbf{u}$ is the sum of a vector and its conjugate, and $\mathbf{w}$ is i times their difference.

Applying $\mathbf{A}$ to $\mathbf{u}$ and $\mathbf{w}$

Since $\mathbf{A}$ is real and linear,

$$\mathbf{A}\mathbf{u} = \mathbf{A}(\mathbf{v} + \bar{\mathbf{v}}) = \mathbf{A}\mathbf{v} + \mathbf{A}\bar{\mathbf{v}} = \lambda\mathbf{v} + \lambda\bar{\mathbf{v}} = \lambda(\mathbf{v} + \bar{\mathbf{v}}) = \lambda\mathbf{u} \quad (20)$$

Similarly,

$$\mathbf{A}\mathbf{w} = \mathbf{A}(i(\mathbf{v} - \bar{\mathbf{v}})) = i(\mathbf{A}\mathbf{v} - \mathbf{A}\bar{\mathbf{v}}) = i(\lambda\mathbf{v} - \lambda\bar{\mathbf{v}}) = \lambda i(\mathbf{v} - \bar{\mathbf{v}}) = \lambda\mathbf{w} \quad (21)$$

Hence we get

$$\mathbf{A}\mathbf{u} = \lambda\mathbf{u} \quad (22)$$

$$\mathbf{A}\mathbf{w} = \lambda\mathbf{w} \quad (23)$$

Real eigenvectors

If $\mathbf{u} \neq \mathbf{0}$, then $\mathbf{u}$ is a real eigenvector corresponding to eigenvalue λ . If $\mathbf{w} \neq \mathbf{0}$, then $\mathbf{w}$ is a real eigenvector corresponding to eigenvalue λ .

Conclusion : For any real symmetric matrix $\mathbf{A}$,

- 1) All eigenvalues of $\mathbf{A}$ are **real**.
- 2) Eigenvectors corresponding to distinct eigenvalues are **orthogonal**.
- 3) For every eigenvalue λ of a real symmetric matrix $\mathbf{A}$, there exists a **real eigenvector**. Hence, the eigenvectors of real symmetric matrices are real.

Case 2 : $\mathbf{A}$ is complex and symmetric

Let $\mathbf{A} \in \mathbb{C}^{n \times n}$ be a complex symmetric matrix satisfying

$$\mathbf{A} = \mathbf{A}^\top \quad (24)$$

Note that $\mathbf{A}$ is not necessarily Hermitian since in general $\mathbf{A} \neq \mathbf{A}^*$.

$$\mathbf{A} \neq \mathbf{A}^* \quad (25)$$

Eigenvalues of $\mathbf{A}$ may be complex

Let $\lambda \in \mathbb{C}$ be an eigenvalue of $\mathbf{A}$ with eigenvector $\mathbf{v} \neq \mathbf{0}$ such that

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \quad (26)$$

Define scalar z as :

$$z = \mathbf{v}^* \mathbf{A} \mathbf{v} \quad (27)$$

Taking complex conjugate transpose on both sides ,

$$z^* = \mathbf{v}^* \mathbf{A}^* \mathbf{v} \quad (28)$$

If $\mathbf{A} = \mathbf{A}^*$, then

$$z = z^* \quad (29)$$

$$z = \bar{z} \text{ as } z \text{ is a scalar} \quad (30)$$

Therefore z is a real scalar value

For Hermitian matrices ($\mathbf{A} = \mathbf{A}^*$), the quantity $\mathbf{v}^* \mathbf{A} \mathbf{v}$ is always real. However, for complex symmetric matrices, $\mathbf{A} \neq \mathbf{A}^*$, so

$$\mathbf{v}^* \mathbf{A} \mathbf{v} \text{ may not be real.} \quad (31)$$

The Rayleigh quotient is

$$\frac{\mathbf{v}^* \mathbf{A} \mathbf{v}}{\mathbf{v}^* \mathbf{v}} = \lambda \quad (32)$$

Since $\mathbf{v}^* \mathbf{A} \mathbf{v}$ may be complex, λ may also be complex.

Eigenvectors corresponding to distinct eigenvalues need not be orthogonal

For complex symmetric matrices, we have only $\mathbf{A} = \mathbf{A}^\top$, so the Hermitian symmetry property does not hold.

Thus, for two eigenpairs $(\lambda_1, \mathbf{v}_1)$ and $(\lambda_2, \mathbf{v}_2)$ with $\lambda_1 \neq \lambda_2$, we **cannot guarantee**

$$\mathbf{v}_1^* \mathbf{v}_2 = 0 \quad (33)$$

Let

$$\mathbf{A} \mathbf{v}_1 = \alpha \mathbf{v}_1 \quad (34)$$

$$\mathbf{A} \mathbf{v}_2 = \beta \mathbf{v}_2 \quad (35)$$

Where $\alpha \neq \beta$ are distinct eigenvalues. Multiplying the first equation on the left by $\mathbf{v}_2^*$:

$$\mathbf{v}_2^* \mathbf{A} \mathbf{v}_1 = \alpha \mathbf{v}_2^* \mathbf{v}_1 \quad (36)$$

As $\mathbf{A} \neq \mathbf{A}^*$ and also $\mathbf{A} \neq \bar{\mathbf{A}}$ we cannot proceed like what we had done earlier for **case 1**.

Hence, eigenvectors corresponding to distinct eigenvalues **need not be orthogonal and are not guaranteed to be real**.

Conclusion : For a complex symmetric matrix $\mathbf{A}$:

- 1) The eigenvalues **can be complex**.
- 2) The eigenvectors corresponding to distinct eigenvalues are **not necessarily orthogonal and need not be real**.

But if $\mathbf{A} = \mathbf{A}^*$ then we get the same result as **case 1**

Examples to Support the Proof

Example 1 : Real and Symmetric Matrix

(a) Let

$$\mathbf{A} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \quad (37)$$

Clearly, $\mathbf{A} = \mathbf{A}^\top$, hence $\mathbf{A}$ is real symmetric.

Finding Eigenvalues

The characteristic equation is:

$$|\mathbf{A} - \lambda \mathbf{I}| = 0 \quad (38)$$

Substituting $\mathbf{A}$,

$$\begin{vmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 2-\lambda \end{vmatrix} = 0 \quad (39)$$

$$\begin{vmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 2-\lambda \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 + \frac{1}{2-\lambda} R_1} \begin{vmatrix} 2-\lambda & -1 & 0 \\ 0 & \frac{(2-\lambda)^2-1}{2-\lambda} & -1 \\ 0 & -1 & 2-\lambda \end{vmatrix} \xrightarrow{R_3 \rightarrow R_3 + \frac{(2-\lambda)}{(2-\lambda)^2-1} R_2} \begin{vmatrix} 2-\lambda & -1 & 0 \\ 0 & \frac{(2-\lambda)^2-1}{2-\lambda} & -1 \\ 0 & 0 & \frac{(2-\lambda)^3-4(2-\lambda)}{(2-\lambda)^2-1} \end{vmatrix} \quad (40)$$

$$(2-\lambda) \left(\frac{(2-\lambda)^2-1}{2-\lambda} \right) \left(\frac{(2-\lambda)^3-4(2-\lambda)}{(2-\lambda)^2-1} \right) = 0 \quad (41)$$

Simplifying,

$$(2-\lambda)((2-\lambda)^2-2) = 0 \quad (42)$$

Hence, the eigenvalues are:

$$\lambda_1 = 2, \lambda_2 = 2 + \sqrt{2}, \lambda_3 = 2 - \sqrt{2} \quad (43)$$

All eigenvalues are real, confirming positive definiteness.

Finding Eigenvectors

For $\lambda_1 = 2$:

$$(\mathbf{A} - 2\mathbf{I})\mathbf{v} = 0 \quad (44)$$

$$\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad (45)$$

$$\left(\begin{array}{ccc|c} 0 & -1 & 0 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \end{array} \right) \xrightarrow{R_2 \leftrightarrow R_1} \left(\begin{array}{ccc|c} -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{array} \right) \xrightarrow{R_3 \rightarrow R_3 - R_2} \left(\begin{array}{ccc|c} -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad (46)$$

Hence, $x = -z, y = 0 \Rightarrow \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$.

For $\lambda_2 = 2 + \sqrt{2}$:

$$(\mathbf{A} - (2 + \sqrt{2})\mathbf{I})\mathbf{v} = 0 \quad (47)$$

$$\begin{pmatrix} -\sqrt{2} & -1 & 0 \\ -1 & -\sqrt{2} & -1 \\ 0 & -1 & -\sqrt{2} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad (48)$$

$$\left(\begin{array}{ccc|c} -\sqrt{2} & -1 & 0 & 0 \\ -1 & -\sqrt{2} & -1 & 0 \\ 0 & -1 & -\sqrt{2} & 0 \end{array} \right) \xleftrightarrow{R_2 \rightarrow R_2 - \frac{1}{\sqrt{2}}R_1} \left(\begin{array}{ccc|c} -\sqrt{2} & -1 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{2}} & -1 & 0 \\ 0 & -1 & -\sqrt{2} & 0 \end{array} \right) \xleftrightarrow{R_3 \rightarrow R_3 - \sqrt{2}R_2} \left(\begin{array}{ccc|c} -\sqrt{2} & -1 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{2}} & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad (49)$$

Simplifying gives proportional relation $y = -\sqrt{2}x$, $z = x$. Hence $\mathbf{v}_2 = \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix}$.

For $\lambda_3 = 2 - \sqrt{2}$: Similarly, solving gives $\mathbf{v}_3 = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}$.

Thus, eigenvectors are **real and mutually orthogonal**.

Example 2 : Complex Symmetric Matrix

(a) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1+i \\ 1+i & 1 \end{pmatrix} \quad (50)$$

Here, $\mathbf{A} = \mathbf{A}^T$ but $\mathbf{A} \neq \mathbf{A}^*$, so $\mathbf{A}$ is complex symmetric but not Hermitian.

Finding Eigenvalues

$$|\mathbf{A} - \lambda \mathbf{I}| = 0 \quad (51)$$

$$\begin{vmatrix} 1-\lambda & 1+i \\ 1+i & 1-\lambda \end{vmatrix} = 0 \quad (52)$$

$$\begin{vmatrix} 1-\lambda & 1+i \\ 1+i & 1-\lambda \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 - \frac{1+i}{1-\lambda} R_1} \begin{vmatrix} 1-\lambda & 1+i \\ 0 & (1-\lambda) - \frac{(1+i)^2}{1-\lambda} \end{vmatrix} \quad (53)$$

$$(1-\lambda)^2 - (1+i)^2 = 0 \quad (54)$$

$$(1-\lambda)^2 - 2i = 0 \quad (55)$$

$$1-\lambda = \pm \sqrt{2i} \Rightarrow \lambda = 1 \mp \sqrt{2i} \quad (56)$$

Hence eigenvalues are:

$$\lambda_1 = 1 - \sqrt{2i}, \quad \lambda_2 = 1 + \sqrt{2i} \quad (57)$$

Finding Eigenvectors

For $\lambda_1 = 1 - \sqrt{2i}$:

$$(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{v} = 0 \quad (58)$$

$$\begin{pmatrix} \sqrt{2i} & 1+i \\ 1+i & \sqrt{2i} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (59)$$

$$\left(\begin{array}{cc|c} \sqrt{2i} & 1+i & 0 \\ 1+i & \sqrt{2i} & 0 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - \frac{1+i}{\sqrt{2i}} R_1} \left(\begin{array}{cc|c} \sqrt{2i} & 1+i & 0 \\ 0 & 0 & 0 \end{array} \right) \quad (60)$$

Hence $y = -\frac{\sqrt{2i}}{1+i}x$ and $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -\frac{\sqrt{2i}}{1+i} \end{pmatrix}$.

For $\lambda_2 = 1 + \sqrt{2i}$:

$$(\mathbf{A} - \lambda_2 \mathbf{I})\mathbf{v} = 0 \quad (61)$$

$$\begin{pmatrix} -\sqrt{2i} & 1+i \\ 1+i & -\sqrt{2i} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (62)$$

$$\left(\begin{array}{cc|c} -\sqrt{2i} & 1+i & 0 \\ 1+i & -\sqrt{2i} & 0 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - \frac{1+i}{-\sqrt{2i}} R_1} \left(\begin{array}{cc|c} -\sqrt{2i} & 1+i & 0 \\ 0 & 0 & 0 \end{array} \right) \quad (63)$$

Hence $y = \frac{\sqrt{2}i}{1+i}x$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ \frac{\sqrt{2}i}{1+i} \end{pmatrix}$.

$$\lambda_1 = 1 - \sqrt{2}i, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ -\frac{\sqrt{2}i}{1+i} \end{pmatrix} \quad (64)$$

$$\lambda_2 = 1 + \sqrt{2}i, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ \frac{\sqrt{2}i}{1+i} \end{pmatrix} \quad (65)$$

The Eigen vectors are **complex**

(b) Let

$$\mathbf{A} = \begin{pmatrix} 1 & i & 0 \\ i & 1 & i \\ 0 & i & 1 \end{pmatrix} \quad (66)$$

Here, $\mathbf{A} = \mathbf{A}^\top$ but $\mathbf{A} \neq \mathbf{A}^*$, so $\mathbf{A}$ is complex symmetric but not Hermitian.

Finding Eigenvalues

$$|\mathbf{A} - \lambda \mathbf{I}| = 0 \quad (67)$$

$$\begin{vmatrix} 1-\lambda & i & 0 \\ i & 1-\lambda & i \\ 0 & i & 1-\lambda \end{vmatrix} = 0 \quad (68)$$

$$\begin{vmatrix} 1-\lambda & i & 0 \\ i & 1-\lambda & i \\ 0 & i & 1-\lambda \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 - \frac{i}{1-\lambda} R_1} \begin{vmatrix} 1-\lambda & i & 0 \\ 0 & \frac{(1-\lambda)^2+1}{1-\lambda} & i \\ 0 & i & 1-\lambda \end{vmatrix} \xrightarrow{R_3 \rightarrow R_3 - \frac{i(1-\lambda)}{(1-\lambda)^2+1} R_2} \begin{vmatrix} 1-\lambda & i & 0 \\ 0 & \frac{(1-\lambda)^2+1}{1-\lambda} & i \\ 0 & 0 & \frac{(1-\lambda)^3+2(1-\lambda)}{(1-\lambda)^2+1} \end{vmatrix} \quad (69)$$

$$(1-\lambda) \left(\frac{(1-\lambda)^2+1}{1-\lambda} \right) \left(\frac{(1-\lambda)^3+2(1-\lambda)}{(1-\lambda)^2+1} \right) = 0 \quad (70)$$

Simplifying,

$$(1-\lambda)((1-\lambda)^2+2) = 0 \quad (71)$$

Hence eigenvalues are:

$$\lambda_1 = 1, \lambda_2 = 1 + i\sqrt{2}, \lambda_3 = 1 - i\sqrt{2} \quad (72)$$

Finding Eigenvectors

For $\lambda_1 = 1$:

$$(\mathbf{A} - \mathbf{I})\mathbf{v} = 0 \quad (73)$$

$$\begin{pmatrix} 0 & i & 0 \\ i & 0 & i \\ 0 & i & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad (74)$$

$$\left(\begin{array}{ccc|c} 0 & i & 0 & 0 \\ i & 0 & i & 0 \\ 0 & i & 0 & 0 \end{array} \right) \xrightarrow{R_2 \leftrightarrow R_1} \left(\begin{array}{ccc|c} i & 0 & i & 0 \\ 0 & i & 0 & 0 \\ 0 & i & 0 & 0 \end{array} \right) \xrightarrow{R_3 \rightarrow R_3 - R_2} \left(\begin{array}{ccc|c} i & 0 & i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad (75)$$

Hence $y = 0, z = -x \Rightarrow \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$.

For $\lambda_2 = 1 + i\sqrt{2}$

$$\mathbf{A} - \lambda_2 \mathbf{I} = \begin{pmatrix} -i\sqrt{2} & i & 0 \\ i & -i\sqrt{2} & i \\ 0 & i & -i\sqrt{2} \end{pmatrix} \quad (76)$$

Forming the augmented matrix :

$$\left(\begin{array}{ccc|c} -i\sqrt{2} & i & 0 & 0 \\ i & -i\sqrt{2} & i & 0 \\ 0 & i & -i\sqrt{2} & 0 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - \frac{i}{-i\sqrt{2}} R_1} \left(\begin{array}{ccc|c} -i\sqrt{2} & i & 0 & 0 \\ 0 & i\left(\frac{1}{\sqrt{2}} - \sqrt{2}\right) & i & 0 \\ 0 & i & -i\sqrt{2} & 0 \end{array} \right) \quad (77)$$

$$\xrightarrow{R_3 \rightarrow R_3 - \frac{i}{i\left(\frac{1}{\sqrt{2}} - \sqrt{2}\right)} R_2} \left(\begin{array}{ccc|c} -i\sqrt{2} & i & 0 & 0 \\ 0 & i\left(\frac{1}{\sqrt{2}} - \sqrt{2}\right) & i & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad (78)$$

Let $x = z$. From the first two equations, $y = \sqrt{2}x$.

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} \quad (79)$$

For $\lambda_3 = 1 - i\sqrt{2}$

$$\mathbf{A} - \lambda_3 \mathbf{I} = \begin{pmatrix} i\sqrt{2} & i & 0 \\ i & i\sqrt{2} & i \\ 0 & i & i\sqrt{2} \end{pmatrix} \quad (80)$$

Forming the augmented matrix:

$$\left(\begin{array}{ccc|c} i\sqrt{2} & i & 0 & 0 \\ i & i\sqrt{2} & i & 0 \\ 0 & i & i\sqrt{2} & 0 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - \frac{i}{i\sqrt{2}} R_1} \left(\begin{array}{ccc|c} i\sqrt{2} & i & 0 & 0 \\ 0 & i\left(\sqrt{2} - \frac{1}{\sqrt{2}}\right) & i & 0 \\ 0 & i & i\sqrt{2} & 0 \end{array} \right) \quad (81)$$

$$\xrightarrow{R_3 \rightarrow R_3 - \frac{i}{i\left(\sqrt{2} - \frac{1}{\sqrt{2}}\right)} R_2} \left(\begin{array}{ccc|c} i\sqrt{2} & i & 0 & 0 \\ 0 & i\left(\sqrt{2} - \frac{1}{\sqrt{2}}\right) & i & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad (82)$$

Hence, $y = -\sqrt{2}x$, $x = z$.

$$\mathbf{v}_3 = \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} \quad (83)$$

Thus, by Gaussian elimination:

$$\lambda_1 = 1, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \quad (84)$$

$$\lambda_2 = 1 + i\sqrt{2}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} \quad (85)$$

$$\lambda_3 = 1 - i\sqrt{2}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} \quad (86)$$

Some of the eigen values are complex but the eigen vectors are **real**.